

Linux Security Audit and Control Assurance Report

Evidence Bundle 1: Auditd configuration and events

Service status

auditd was installed, enabled at boot and started. It is loaded, enabled and active (running) since 12:46:41 on 2 October 2026 (PID 11423).

```
Preparing to unpack ./libauplugi1:amd64 (1:4.1.2-1ubuntu0.1_amd64.deb)...
Unpacking libauplugi1:amd64 (1:4.1.2-1ubuntu0.1)...
Selecting previously unselected package auditd.
Preparing to unpack ./auditd_1:4.1.2-1ubuntu0.1_amd64.deb...
Unpacking auditd (1:4.1.2-1ubuntu0.1)...
Selecting previously unselected package audispd-plugins.
Preparing to unpack ./audispd-plugins_1:4.1.2-1ubuntu0.1_amd64.deb...
Unpacking audispd-plugins (1:4.1.2-1ubuntu0.1)...
Setting up libauparse0t64:amd64 (1:4.1.2-1ubuntu0.1)...
Setting up libauplugi1:amd64 (1:4.1.2-1ubuntu0.1)...
Setting up auditd (1:4.1.2-1ubuntu0.1)...
Created symlink '/etc/systemd/system/multi-user.target.wants/audit-rules.service'
' -> '/usr/lib/systemd/system/audit-rules.service'.
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' ->
'/usr/lib/systemd/system/auditd.service'.
Setting up audispd-plugins (1:4.1.2-1ubuntu0.1)...
Processing triggers for man-db (2.13.1-1build1)...
Processing triggers for libc-bin (2.43-2ubuntu2)...
Synchronizing state of auditd.service with SysV service script with /usr/lib/systemd/systemd-sysv
Executing: /usr/lib/systemd/systemd-sysv-install enable auditd
● auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; enabled; preset: enabled)
   Active: active (running) since Fri 2026-10-02 12:46:41 WAT; 3s ago
 Invocation: f85021a7f2234afcaeb8e645907bf95b
    Docs: man:auditd(8)
          https://github.com/linux-audit/audit-documentation
 Main PID: 11423 (auditd)
   Tasks: 2 (limit: 7380)
  Memory: 508K (peak: 1.9M)
     CPU: 11ms
    CGroup: /system.slice/auditd.service
            └─11423 /usr/sbin/auditd

Oct 02 12:46:41 funke-akindele-VNware-Virtual-Platform systemd[1]: Starting au...
Oct 02 12:46:41 funke-akindele-VNware-Virtual-Platform auditd[11423]: No plugi...
Oct 02 12:46:41 funke-akindele-VNware-Virtual-Platform auditd[11423]: Init com...
Oct 02 12:46:41 funke-akindele-VNware-Virtual-Platform systemd[1]: Started aud...
Hint: Some lines were ellipsized, use -l to show in full.
```

Figure 1. auditd service status.

Loaded audit rules

Rules were written to /etc/audit/rules.d/custom.rules, loaded with augenrules --load, and listed with auditctl -l.

Rule	Purpose
-w /etc/passwd -p rwx -k passwd_changes	Watch the account list
-w /etc/shadow -p rwx -k shadow_changes	Watch the password hash file
-a always,exit -F arch=b64 -S execve -k program_execution	Record program starts (64 bit)
-a always,exit -F arch=b32 -S execve -k program_execution	Record program starts (32 bit)
-w /var/log/auth.log -p wa -k auth_failures	Watch changes to the authentication log

```

EOF
[ -f /var/log/auth.log ] && echo "-w /var/log/auth.log -p wa -k auth_failures" | sudo tee -a /etc
sudo augenrules --load; sudo systemctl restart auditd; sudo auditctl -l
No rules
enabled 1
failure 1
pid 11423
rate_limit 0
backlog_limit 8192
lost 0
backlog 4
backlog_wait_time 60000
backlog_wait_time_actual 0
enabled 1
failure 1
pid 11423
rate_limit 0
backlog_limit 8192
lost 0
backlog 4
backlog_wait_time 60000
backlog_wait_time_actual 0
enabled 1
failure 1
pid 11423
rate_limit 0
backlog_limit 8192
lost 0
backlog 4
backlog_wait_time 60000
backlog_wait_time_actual 0
Old style watch rules are slower
Old style watch rules are slower
-w /etc/passwd -p rwx -k passwd_changes
-w /etc/shadow -p rwx -k shadow_changes
-a always,exit -F arch=b64 -S execve -F key=program_execution
-a always,exit -F arch=b32 -S execve -F key=program_execution
-w /var/log/auth.log -p wa -k auth_failures

```

Figure 2. Output of auditctl -l (five rules loaded).

ausearch evidence

Time (2 Oct 2026)	User (audit)	Action	Key
12:53:19	funke-akindele	Ran auditctl through sudo and added the rules (add_rule)	auth_failures
12:53:20	unset (system)	Rules removed and added again when the service restarted	auth_failures
12:54:03	funke-akindele	Ran /usr/sbin/ausearch as root from terminal pts1	program_execution

```

type=PROCTITLE msg=audit(10/02/2026 12:54:03.069:620) : proctitle=ausearch -k program_execution -l
type=PATH msg=audit(10/02/2026 12:54:03.069:620) : item=1 name=/lib64/ld-linux-x86-64.so.2 inode=150193 dev=08:02 mode=file,755 ouid=root ogid=root rd
ev=00:00 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=PATH msg=audit(10/02/2026 12:54:03.069:620) : item=0 name=/usr/sbin/ausearch inode=143595 dev=08:02 mode=file,755 ouid=root ogid=root rdev=00:00
nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(10/02/2026 12:54:03.069:620) : cwd=/home/funke-akindele
type=EXECVE msg=audit(10/02/2026 12:54:03.069:620) : argc=4 a0=ausearch a1=-k a2=program_execution a3=-i
type=SYSCALL msg=audit(10/02/2026 12:54:03.069:620) : arch=x86_64 syscall=execve success=yes exit=0 a0=0x572fb4134e00 a1=0x572fb4137c50 a2=0x572fb4150
3a0 a3=0x0 items=2 ppid=12608 pid=12669 auid=funke-akindele uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts1 s
es=3 comm=ausearch exe=/usr/sbin/ausearch subj=unconfined key=program_execution
=== auth_failures ===
....
type=PROCTITLE msg=audit(10/02/2026 12:53:19.944:464) : proctitle=/sbin/auditctl -R /etc/audit/audit.rules
type=PATH msg=audit(10/02/2026 12:53:19.944:464) : item=0 name=/var/log/ inode=1835017 dev=08:02 mode=dir,775 ouid=root ogid=syslog rdev=00:00 nametyp
e=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(10/02/2026 12:53:19.944:464) : cwd=/home/funke-akindele
type=SOCKADDR msg=audit(10/02/2026 12:53:19.944:464) : saddr={ saddr_fam=netlink nlnk_fam=16 nlnk_pid=0 }
type=SYSCALL msg=audit(10/02/2026 12:53:19.944:464) : arch=x86_64 syscall=sendto success=yes exit=1088 a0=0x3 a1=0x7fffd3f7f56e0 a2=0x440 a3=0x0 items=
1 ppid=12535 pid=12555 auid=funke-akindele uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=pts1 ses=3 comm=auditctl e
xe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(10/02/2026 12:53:19.944:464) : auid=funke-akindele ses=3 subj=unconfined op=add_rule key=auth_failures list=exit res=yes
....
type=PROCTITLE msg=audit(10/02/2026 12:53:20.076:528) : proctitle=/sbin/auditctl -R /etc/audit/audit.rules
type=SOCKADDR msg=audit(10/02/2026 12:53:20.076:528) : saddr={ saddr_fam=netlink nlnk_fam=16 nlnk_pid=0 }
type=SYSCALL msg=audit(10/02/2026 12:53:20.076:528) : arch=x86_64 syscall=sendto success=yes exit=1088 a0=0x3 a1=0x7ffc82bb0610 a2=0x440 a3=0x0 items=
0 ppid=12574 pid=12595 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=auditctl e
xe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(10/02/2026 12:53:20.076:528) : auid=unset ses=unset subj=unconfined op=remove_rule key=auth_failures list=exit res=yes
....
type=PROCTITLE msg=audit(10/02/2026 12:53:20.081:536) : proctitle=/sbin/auditctl -R /etc/audit/audit.rules
type=PATH msg=audit(10/02/2026 12:53:20.081:536) : item=0 name=/var/log/ inode=1835017 dev=08:02 mode=dir,775 ouid=root ogid=syslog rdev=00:00 nametyp
e=PARENT cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(10/02/2026 12:53:20.081:536) : cwd=/
type=SOCKADDR msg=audit(10/02/2026 12:53:20.081:536) : saddr={ saddr_fam=netlink nlnk_fam=16 nlnk_pid=0 }
type=SYSCALL msg=audit(10/02/2026 12:53:20.081:536) : arch=x86_64 syscall=sendto success=yes exit=1088 a0=0x3 a1=0x7ffc82bb2ad0 a2=0x440 a3=0x0 items=
1 ppid=12574 pid=12595 auid=unset uid=root gid=root euid=root suid=root fsuid=root egid=root sgid=root fsgid=root tty=(none) ses=unset comm=auditctl e
xe=/usr/sbin/auditctl subj=unconfined key=(null)
type=CONFIG_CHANGE msg=audit(10/02/2026 12:53:20.081:536) : auid=unset ses=unset subj=unconfined op=add_rule key=auth_failures list=exit res=yes

```

Figure 3. ausearch results for program_execution and auth_failures.

aureport evidence

Report period 12:46:41 to 12:55:01: 0 failed logins, 0 failed authentications, 0 account changes, 2 failed system calls, 2 AVC events, 3 events in total. The login report found no events of interest.

```
=== FAILED ===
Failed Summary Report
=====
Range of time in logs: 10/02/2026 12:46:41.753 - 10/02/2026 12:55:01.246
Selected time for report: 10/02/2026 12:46:41 - 10/02/2026 12:55:01.246
Number of changes in configuration: 0
Number of logins: 0
Number of failed logins: 0
Number of authentications: 0
Number of failed authentications: 0
Number of users: 1
Number of terminals: 2
Number of host names: 1
Number of executables: 2
Number of commands: 2
Number of files: 0
Number of AVC's: 2
Number of MAC events: 0
Number of failed syscalls: 2
Number of anomaly events: 0
Number of responses to anomaly events: 0
Number of crypto events: 0
Number of integrity events: 0
Number of virt events: 0
Number of keys: 0
Number of process IDs: 2
Number of events: 3

=== LOGIN ===
Login Report
=====
# date time auid host term exe success event
=====
<no events of interest were found>
```

Figure 4. aureport --failed and aureport --login.

Interpretation

User funke-akindele (login ID 1000) loaded the audit rules through sudo at 12:53:19 and ran ausearch as root at 12:54:03. The auid field keeps the name of the original login user even after sudo, so the record shows who acted and when. This matters for governance because management can prove who changed a sensitive control, and changes to audit rules are a common way to hide activity. The entry at 12:53:20 with auid unset is the system reloading its rules after the restart, which is normal.

Limits: the auth_failures key matched the loading of the rule, not failed logins. Auditd started only at 12:46:41, so the 0 failed logins covers a short period. The 2 AVC events and 2 failed system calls were not examined.

Evidence Bundle 2: Linux log analysis

Review	Result
journalctl evidence	Errors from journalctl -p err: repeated chronyd "NTS-KE session timed out" messages for the Ubuntu time servers between 12:31 and 12:40, and "Failed to start wazuh-indexer.service" at 12:46:41. The 10 second live view showed only the lab sudo commands.
Authentication and privilege use	auth.log covers 12:46:44 to 12:56:14. The "failed password" search matched only the grep command itself, so there were no real failed logins in this file. All sudo entries are by funke-akindele to root and match lab steps.

Review	Result
Second source: ssh journal	Failed passwords for account attacksim from 100.65.116.44, dated 18 May (year not shown). At least 8 sshd failures from 12:56:53 to 12:57:25 (32 seconds), each followed by a unix_chkpwd failure, from three source ports, so at least three connections. Output was limited to 15 lines, so the true count may be higher.
Error and warning analysis	Taken from the journal, which is the equivalent source here. Two conditions matter: the failed monitoring service and the failed time sync.

```

Oct 02 12:31:43 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.123:4460 (2.ntp.ubuntu.com) timed out
Oct 02 12:32:31 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.123:4460 (1.ntp.ubuntu.com) timed out
Oct 02 12:32:31 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.123:4460 (ntp-bootstrap.ubuntu.com) timed out
Oct 02 12:33:03 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.113:4460 (4.ntp.ubuntu.com) timed out
Oct 02 12:33:03 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.112:4460 (3.ntp.ubuntu.com) timed out
Oct 02 12:33:04 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.111:4460 (ntp-bootstrap.ubuntu.com) timed out
Oct 02 12:33:04 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.123:4460 (2.ntp.ubuntu.com) timed out
Oct 02 12:34:56 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.122:4460 (1.ntp.ubuntu.com) timed out
Oct 02 12:34:57 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.121:4460 (ntp-bootstrap.ubuntu.com) timed out
Oct 02 12:35:29 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.113:4460 (4.ntp.ubuntu.com) timed out
Oct 02 12:35:29 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.112:4460 (3.ntp.ubuntu.com) timed out
Oct 02 12:35:30 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.111:4460 (ntp-bootstrap.ubuntu.com) timed out
Oct 02 12:35:30 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.123:4460 (2.ntp.ubuntu.com) timed out
Oct 02 12:39:29 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.122:4460 (1.ntp.ubuntu.com) timed out
Oct 02 12:39:30 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.121:4460 (ntp-bootstrap.ubuntu.com) timed out
Oct 02 12:40:02 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.113:4460 (4.ntp.ubuntu.com) timed out
Oct 02 12:40:02 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.112:4460 (3.ntp.ubuntu.com) timed out
Oct 02 12:40:03 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 91.189.91.111:4460 (ntp-bootstrap.ubuntu.com) timed out
Oct 02 12:40:03 funke-akindele-VMware-Virtual-Platform chronyd[1254]: NTP-KE session with 185.125.190.123:4460 (2.ntp.ubuntu.com) timed out
Oct 02 12:46:41 funke-akindele-VMware-Virtual-Platform systemd[1]: Failed to start wazuh-indexer.service - wazuh-indexer.
== LIVE VIEW 10 SECONDS ==
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12735]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/journalctl -u ssh --no-pager -n 15
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12735]: pam_unix(sudo:session): session closed for user root
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12739]: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12739]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/journalctl --since today --no-pager
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12739]: pam_unix(sudo:session): session closed for user root
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12744]: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12744]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/journalctl -p err --no-pager -n 20
Oct 02 12:55:38 funke-akindele-VMware-Virtual-Platform sudo[12744]: pam_unix(sudo:session): session closed for user root
Oct 02 12:55:39 funke-akindele-VMware-Virtual-Platform sudo[12748]: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
Oct 02 12:55:39 funke-akindele-VMware-Virtual-Platform sudo[12748]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/timedatectl -f --no-pager

```

Figure 5. Journal errors and live view.

```

usr/bin/timedatectl -f --no-pager
funke-akindele@funke-akindele-VMware-Virtual-Platform:~$ echo "=== FAILED PASSWORD ==="; sudo grep -i "failed password" /var/log/auth.log | tail -n 15
|| true
echo "=== SUDO ==="; sudo grep -i "sudo" /var/log/auth.log | tail -n 15
echo "=== FIRST AND LAST LINE OF AUTH.LOG ==="; sudo head -n 1 /var/log/auth.log; sudo tail -n 1 /var/log/auth.log
=== FAILED PASSWORD ===
2026-10-02T12:56:14.778945+01:00 funke-akindele-VMware-Virtual-Platform sudo: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/grep -i failed password /var/log/auth.log
=== SUDO ===
2026-10-02T12:55:38.820318+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T12:55:38.829617+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
2026-10-02T12:55:38.829842+01:00 funke-akindele-VMware-Virtual-Platform sudo: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/journalctl --since today --no-pager
2026-10-02T12:55:38.941235+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T12:55:38.955605+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
2026-10-02T12:55:38.955797+01:00 funke-akindele-VMware-Virtual-Platform sudo: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/journalctl -p err --no-pager -n 20
2026-10-02T12:55:38.998558+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T12:55:39.008979+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
2026-10-02T12:55:39.001228+01:00 funke-akindele-VMware-Virtual-Platform sudo: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/timedatectl -f --no-pager
2026-10-02T12:55:49.084977+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T12:56:14.778712+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
2026-10-02T12:56:14.778945+01:00 funke-akindele-VMware-Virtual-Platform sudo: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/grep -i failed password /var/log/auth.log
2026-10-02T12:56:14.781528+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T12:56:14.792794+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
2026-10-02T12:56:14.792940+01:00 funke-akindele-VMware-Virtual-Platform sudo: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/grep -i sudo /var/log/auth.log
=== FIRST AND LAST LINE OF AUTH.LOG ===
2026-10-02T12:46:44.441700+01:00 funke-akindele-VMware-Virtual-Platform sudo: pam_unix(sudo:session): session closed for user root
2026-10-02T12:56:14.829665+01:00 funke-akindele-VMware-Virtual-Platform sudo: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/tail -n 1 /var/log/auth.log

```

Figure 6. auth.log search for failed password and sudo entries.

```

1000)
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12803]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/grep -i sudo /var/log/auth.log
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12803]: pam_unix(sudo:session): session closed for user root
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12808]: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12808]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/grep -i sudo /var/log/auth.log
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12808]: pam_unix(sudo:session): session closed for user root
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12813]: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12813]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/head -n 1 /var/log/auth.log
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12813]: pam_unix(sudo:session): session closed for user root
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12822]: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12822]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/tail -n 1 /var/log/auth.log
Oct 02 12:56:14 funke-akindele-VMware-Virtual-Platform sudo[12822]: pam_unix(sudo:session): session closed for user root
Oct 02 12:56:45 funke-akindele-VMware-Virtual-Platform sudo[12847]: pam_unix(sudo:session): session opened for user root(uid=0) by funke-akindele(uid=1000)
Oct 02 12:56:45 funke-akindele-VMware-Virtual-Platform sudo[12847]: funke-akindele : TTY=/dev/pts/0 ; PWD=/home/funke-akindele ; USER=root ; COMMAND=/usr/bin/journalctl -COMM=sudo --no-pager
May 18 12:56:53 funke-akindele-VMware-Virtual-Platform sshd-session[6557]: Failed password for attacksin from 100.65.116.44 port 41996 ssh2
May 18 12:56:56 funke-akindele-VMware-Virtual-Platform unix_chkpwd[7783]: password check failed for user (attacksin)
May 18 12:56:58 funke-akindele-VMware-Virtual-Platform sshd-session[7459]: Failed password for attacksin from 100.65.116.44 port 33354 ssh2
May 18 12:57:00 funke-akindele-VMware-Virtual-Platform unix_chkpwd[8517]: password check failed for user (attacksin)
May 18 12:57:02 funke-akindele-VMware-Virtual-Platform sshd-session[7459]: Failed password for attacksin from 100.65.116.44 port 33354 ssh2
May 18 12:57:03 funke-akindele-VMware-Virtual-Platform unix_chkpwd[8582]: password check failed for user (attacksin)
May 18 12:57:05 funke-akindele-VMware-Virtual-Platform sshd-session[7459]: Failed password for attacksin from 100.65.116.44 port 33354 ssh2
May 18 12:57:06 funke-akindele-VMware-Virtual-Platform unix_chkpwd[8667]: password check failed for user (attacksin)
May 18 12:57:08 funke-akindele-VMware-Virtual-Platform sshd-session[7459]: Failed password for attacksin from 100.65.116.44 port 33354 ssh2
May 18 12:57:08 funke-akindele-VMware-Virtual-Platform unix_chkpwd[8724]: password check failed for user (attacksin)
May 18 12:57:11 funke-akindele-VMware-Virtual-Platform sshd-session[7459]: Failed password for attacksin from 100.65.116.44 port 33354 ssh2
May 18 12:57:15 funke-akindele-VMware-Virtual-Platform unix_chkpwd[8921]: password check failed for user (attacksin)
May 18 12:57:17 funke-akindele-VMware-Virtual-Platform sshd-session[7459]: Failed password for attacksin from 100.65.116.44 port 33354 ssh2
May 18 12:57:23 funke-akindele-VMware-Virtual-Platform unix_chkpwd[10187]: password check failed for user (attacksin)
May 18 12:57:25 funke-akindele-VMware-Virtual-Platform sshd-session[9322]: Failed password for attacksin from 100.65.116.44 port 49184 ssh2

```

Figure 7. ssh journal showing failed passwords for attacksim.

Event table

Event or condition	Evidence source	Time	User or service	Security significance	Recommended action
Burst of failed ssh passwords for attacksim from 100.65.116.44	ssh journal	18 May, 12:56:53 to 12:57:25	sshd, attacksim	Looks like password guessing. Account may be a test account.	Confirm account and source. Check for accepted logins. Install fail2ban.
wazuh-indexer failed to start	journalctl -p err	2 Oct, 12:46:41	systemd	Monitoring data store down. Events may be lost.	Fix the service and add a failure alert.
Time sync timeouts	journalctl -p err	2 Oct, 12:31 to 12:40	chronyd	Wrong clock weakens audit evidence.	Check chronyc tracking. Open port 4460 or use an internal time source.
Sudo to root	auth.log	2 Oct, 12:46 to 12:56	funke-akindele	Privileged use. All matched lab steps.	None. Keep the record.
No failed passwords in current auth.log	auth.log	From 12:46:44	Not applicable	Clean, but under one hour of history.	Keep logs longer and send them to a central store.

Evidence Bundle 3: Lynis security assessment

Item	Result
Version and command	Lynis 3.1.6, run with sudo lynis audit system --no-colors
Hardening index	64 out of 100
Warnings	2: PKGS-7392 (vulnerable packages) and NETW-3015 (promiscuous interface ens33)
Suggestions	At least 15 (first 15 shown in the evidence)
Report and log files	/var/log/lynis-report.dat and /var/log/lynis.log (written 2 Oct 13:00)

```

Enterprise support available (compliance, plugins, interface and tools)

=====
[TIP]: Enhance Lynis audits by adding your settings to custom.prf (see /etc/lynis/default.prf for all settings)

funke-akindele@funke-akindele-Virtual-Platform:~$ echo "=== VERSION ==="; lynis --version
echo "=== HARDENING INDEX ==="; sudo grep "hardening_index" /var/log/lynis-report.dat
echo "=== WARNINGS ==="; sudo grep "warning\[|\]" /var/log/lynis-report.dat
echo "=== SUGGESTIONS (first 15) ==="; sudo grep "suggestion\[|\]" /var/log/lynis-report.dat | head -n 15
echo "=== FILE LOCATIONS ==="; ls -l /var/log/lynis.log /var/log/lynis-report.dat
=== VERSION ===
3.1.6
=== HARDENING INDEX ===
hardening_index=64
=== WARNINGS ===
warning[PKGS-7392]Found one or more vulnerable packages.|-|-|
warning[NETW-3015]Found promiscuous interface|ens33|text:Determine if this mode is required or whitelist interface in profile|
=== SUGGESTIONS (first 15) ===
suggestion[DEB-0280]Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions|-|-|
suggestion[DEB-0810]Install apt-listbugs to display a list of critical bugs prior to each APT installation.|-|-|
suggestion[DEB-0811]Install apt-listchanges to display any significant changes prior to any upgrade via APT.|-|-|
suggestion[DEB-0831]Install needrestart, alternatively to debian-goodies, so that you can run needrestart after upgrades to determine which daemons
are using old versions of libraries and need restarting.|-|-|
suggestion[DEB-0880]Install fail2ban to automatically ban hosts that commit multiple authentication errors.|-|-|
suggestion[BOOT-5122]Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode without password)|-|-|
suggestion[BOOT-5180]Determine runlevel and services at startup|-|-|
suggestion[BOOT-5264]Consider hardening system services|Run '/usr/bin/systemd-analyze security SERVICE' for each service|-|-|
suggestion[AUTH-9229]Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values|-|-|
suggestion[AUTH-9230]Configure password hashing rounds in /etc/login.defs|-|-|
suggestion[AUTH-9282]When possible set expire dates for all password protected accounts|-|-|
suggestion[AUTH-9286]Configure minimum password age in /etc/login.defs|-|-|
suggestion[AUTH-9286]Configure maximum password age in /etc/login.defs|-|-|
suggestion[AUTH-9328]Default umask in /etc/login.defs could not be found and defaults usually to 022, which could be more strict like 027|-|-|
suggestion[FILE-6310]To decrease the impact of a full /home file system, place /home on a separate partition|-|-|
=== FILE LOCATIONS ===
-rw-r----- 1 root root 93525 Oct  2 13:00 /var/log/lynis-report.dat
-rw-r----- 1 root root 779947 Oct  2 13:00 /var/log/lynis.log

```

Figure 8. Lynis version, hardening index, warnings, suggestions and file locations.

Findings and prioritised recommendations

Finding	Risk or control issue	Likely owner	Recommendation	Priorit y
PKGS-7392 vulnerable packages	Known flaws not patched	System administrator	apt update and apt upgrade, restart if needed, rerun Lynis	High
DEB-0880 fail2ban not installed	Repeated failed logins are not blocked (see attacksim)	Security operations and system administrator	Install fail2ban with an sshd rule	High
NETW-3015 promiscuous interface ens33	Card may be reading all traffic	Network team	Check with ip link show ens33. Turn off if not needed.	Mediu m
BOOT-5122 no GRUB password	Boot settings can be changed at the console	System administrator	Set a GRUB password	Mediu m

Finding	Risk or control issue	Likely owner	Recommendation	Priority
AUTH-9286 no password age limits	Old passwords stay valid	Identity and access owner	Set limits in /etc/login.defs	Medium
AUTH-9328 default umask	New files may be too open	System administrator	Set umask 027 in /etc/login.defs	Low

Hardening action

Step	Detail
Before	permitrootlogin prohibit-password (key login for root allowed)
Change	Backup saved as sshd_config.bak. PermitRootLogin no written to /etc/ssh/sshd_config.d/99-hardening.conf. sshd -t passed and ssh was restarted.
After	permitrootlogin no
Objective and impact	No direct root login, so each admin action is tied to a named person. Admins must use their own account and sudo. Jobs that log in as root will stop working.
Lynis retest	Started after the change, but the result was not captured, so no new index is reported.

```

echo "=== FILE LOCATIONS ==="; ls -l /var/log/lynis.log /var/log/lynis-report.dat
=== VERSION ===
3.1.6
=== HARDENING INDEX ===
hardening_index=64
=== WARNINGS ===
warning[]=PKGS-7392[Found one or more vulnerable packages.|-|-|]
warning[]=NETW-3015[Found promiscuous interface|ens33|text:Determine if this mode is required or whitelist interface in profile]
=== SUGGESTIONS (First 15) ===
suggestion[]=DEB-0280[Install libpam-tmpdir to set $TMP and $TMPDIR for PAM sessions.|-|-|]
suggestion[]=DEB-0810[Install apt-listbugs to display a list of critical bugs prior to each APT installation.|-|-|]
suggestion[]=DEB-0811[Install apt-listchanges to display any significant changes prior to any upgrade via APT.|-|-|]
suggestion[]=DEB-0831[Install needrestart, alternatively to debian-goodies, so that you can run needrestart after upgrades to determine which daemons
are using old versions of libraries and need restarting.|-|-|]
suggestion[]=DEB-0880[Install fail2ban to automatically ban hosts that commit multiple authentication errors.|-|-|]
suggestion[]=BOOT-5122[Set a password on GRUB boot loader to prevent altering boot configuration (e.g. boot in single user mode without password)|-|-|]
suggestion[]=BOOT-5180[Determine runlevel and services at startup.|-|-|]
suggestion[]=BOOT-5264[Consider hardening system services|Run '/usr/bin/systemd-analyze security SERVICE' for each service.|-|-|]
suggestion[]=AUTH-9229[Check PAM configuration, add rounds if applicable and expire passwords to encrypt with new values.|-|-|]
suggestion[]=AUTH-9230[Configure password hashing rounds in /etc/login.defs.|-|-|]
suggestion[]=AUTH-9282[When possible set expire dates for all password protected accounts.|-|-|]
suggestion[]=AUTH-9286[Configure minimum password age in /etc/login.defs.|-|-|]
suggestion[]=AUTH-9286[Configure maximum password age in /etc/login.defs.|-|-|]
suggestion[]=AUTH-9328[Default umask in /etc/login.defs could not be found and defaults usually to 022, which could be more strict like 027.|-|-|]
suggestion[]=FILE-6310[To decrease the impact of a full /home file system, place /home on a separate partition.|-|-|]
=== FILE LOCATIONS ===
-rw-r----- 1 root root 93525 Oct  2 13:00 /var/log/lynis-report.dat
-rw-r----- 1 root root 779947 Oct  2 13:00 /var/log/lynis.log
funke@kindleofunke-ubuntu:~$ echo "=== BEFORE ==="; sudo sshd -T | grep -i permitrootlogin
sudo cp /etc/ssh/sshd_config /etc/ssh/sshd_config.bak
echo "PermitRootLogin no" | sudo tee /etc/ssh/sshd_config.d/99-hardening.conf > /dev/null
sudo sshd -t && sudo systemctl restart ssh
echo "=== AFTER ==="; sudo sshd -T | grep -i permitrootlogin
sudo lynis audit system --no-colors > /dev/null 2>&1; sudo grep "hardening_index" /var/log/lynis-report.dat
=== BEFORE ===
permitrootlogin prohibit-password
=== AFTER ===
permitrootlogin no

```

Figure 9. permitrootlogin before and after the change.

Evidence Bundle 4: Control monitoring and governance

Control monitoring table

Control or objective	Evidence	Owner	Observed status	KPI or KRI threshold	Risk or significance	Remediation	Retest or follow-up
Audit service running	auditd status (Figure 1)	System administrator	Active and enabled since 12:46:41	100% of servers running. Alert if stopped over 5 minutes.	No audit trail if stopped	Keep enabled at boot, alert on stop	Monthly script check
Audit rule coverage	auditctl -l (Figure 2)	Security operations	5 rules loaded	All required rules on every server	Missing rules hide changes	Keep rules file under change control	Quarterly and after any change
Failed login monitoring	ssh journal (Figure 7)	Security operations	8 or more failures on attacksim in 32 seconds. None in current logs.	5 failures from one source in 10 minutes raises an alert	Password guessing, no automatic block	Confirm source, install fail2ban	Test failed logins from a test host and confirm the ban
Privileged use	auth.log sudo (Figure 6)	Line manager and IT security	One user, all uses explained	0 unexplained sudo events	Misuse of high rights	Match to tickets, remove rights not needed	Quarterly access review
Patch status	Lynis PKGS-7392 (Figure 8)	System administrator	Vulnerable packages found	Critical in 14 days, others in 30	Known flaws can be used	apt upgrade and restart	Rerun Lynis, then monthly
Configuration hardening	Lynis index, sshd -T (Figures 8, 9)	System owner	Index 64, 2 warnings. Root login set to no.	Index 80 or more, no high finding open over 30 days	Weak settings give easy routes	Fix by priority, record exceptions	Rerun Lynis monthly and compare

Control or objective	Evidence	Owner	Observed status	KPI or KRI threshold	Risk or significance	Remediation	Retest or follow-up
Monitoring platform health	Journal error (Figure 5)	Platform or SOC team	wazuh-indexer failed to start	100% of monitoring services up. Alert within 15 minutes.	Events may not be stored	Fix service, add failure alert	Daily check, then send a test event
Time synchronisation	chronyd errors (Figure 5)	System administrator	Timeouts from 12:31 to 12:40	Clock offset under 1 second	Wrong clock weakens evidence	Fix path to time servers	chronyc tracking weekly

Evidence-backed governance issues

Issue	Evidence	Why it needs governance attention	Owner	Escalate to
Repeated failed ssh logins with no blocking control	Figure 7 and Lynis DEB-0880 (Figure 8)	A preventive control is missing and the activity was seen only in the journal, not by auditd	Security operations	CISO if logins continue or any login succeeds
Monitoring platform not running	Figure 5 (wazuh-indexer failed at 12:46:41)	The detective control that should store events is down, so assurance cannot rely on it	Platform or SOC team	CISO if down over 24 hours
Known vulnerable packages left unpatched	Lynis PKGS-7392 (Figure 8)	A patching control weakness that stays open until a fix date is met	System administrator	Risk function if open over 30 days
Audit coverage starts only at 12:46:41	Figures 1 and 6	No audit history before today and under one hour of auth.log. Evidence is not retained long enough.	Security operations	Risk function until retention is set

Evidence Bundle 5: SIEM and automation mapping

Flow: host evidence is collected by an agent or log forwarder, sent to the SIEM collector, parsed into standard fields, matched against rules, raised as an alert, shown on a dashboard, and summarised in the monthly control report.

Linux evidence	How it reaches the SIEM	Fields used	Monitoring rule	Stays an operational alert	Becomes a governance issue
auditd events (/var/log/audit/audit.log)	audisp plugin or agent	Time, auid, uid, exe, key, result	Hit on passwd_changes or shadow_changes with no ticket. auditd stopped.	One event that matches a ticket	Change with no ticket. auditd off or rules missing on a key server.
auth.log and ssh journal	rsyslog or agent	Time, user, source address, result	5 failures from one source in 10 minutes. Success after failures.	Short burst that is blocked	Bursts continue over days, privileged accounts targeted, no blocking control
Journal errors (systemd, chronyd)	Journal forwarding	Unit, priority, message	Monitoring service failed. Time sync failed.	One service restart that clears the fault	Monitoring platform down over 24 hours. Clock not synced.
Lynis report (lynis-report.dat)	Scheduled copy to a central store	Index, warning and suggestion IDs, date	Index below 80. High finding open over 30 days.	New low priority suggestion	Index falls on two scans. High finding past its fix date.
Configuration checks (for example sshd -T)	Scheduled script saving dated output	Host, setting, value	PermitRootLogin is not no	Drift corrected the same day	Repeated drift, or exceptions with no approval

Evidence Bundle 6: Final audit report

Executive summary

The Linux system was reviewed on 2 October 2026. The audit service is running with five rules loaded, and the Lynis hardening index is 64 out of 100 with 2 warnings. Four conditions need management attention: failed ssh password attempts against the account attacksim with no blocking control, unpatched vulnerable packages, a monitoring service (wazuh-indexer) that failed to start, and audit evidence that begins only on the day of review. Direct root login over ssh was switched off during the review. Overall, the key controls exist but are not yet reliable enough to give assurance, mainly because evidence is short and monitoring is not confirmed to be working.

Scope and methods

Scope: one Ubuntu virtual machine, covering system auditing, authentication and system logs, and a configuration assessment. Methods: auditd was installed and given rules, events were created and searched with ausearch and aureport, logs were read with journalctl and grep, and Lynis 3.1.6 was run. Findings were linked to owners, thresholds and a retest plan. Screenshots are in Bundles 1 to 3.

Findings, priorities, recommendations and owners

Ref	Finding	Evidence	Priority	Recommendation	Owner	Target
F1	Failed ssh passwords on attacksim, no blocking	Figure 7	High	Confirm account and source, check for accepted logins, install fail2ban	Security operations and system administrator	7 days
F2	Vulnerable packages (PKGS-7392)	Figure 8	High	Patch, restart, rerun Lynis	System administrator	14 days
F3	wazuh-indexer failed to start	Figure 5	High	Fix the service, add a failure alert	Platform or SOC team	7 days
F4	Audit and log history starts at 12:46	Figures 1, 6	Medium	Set log retention and forward logs to a central store	Security operations	30 days
F5	Time sync timeouts	Figure 5	Medium	Fix path to time servers	System administrator and network team	30 days
F6	Promiscuous interface ens33	Figure 8	Medium	Find cause, turn off if not needed	Network team	30 days
F7	No GRUB password, no password age limits, umask	Figure 8	Low to medium	Set GRUB password and limits in /etc/login.defs	System administrator and identity owner	60 days

Ref	Finding	Evidence	Priority	Recommendation	Owner	Target
F8	Root login over ssh	Figure 9	Closed	Set to no. Keep under monthly check.	System administrator	Done

Order of work: F1, F2 and F3 first, because they affect attack exposure and the ability to detect events. The system owner accepts any risk left open after the target date.

Escalation triggers

Escalate to	Trigger
CISO	auditd off or rules missing on a key system. Account file changes with no ticket. Failed logins that continue after a block, or any success after a burst. Monitoring down over 24 hours.
Risk function	A high finding open over 30 days. The same finding on two scans. Hardening index falling on two scans.
Management committee	Same threshold missed on several systems. A fix date missed twice. Exposure of a critical business service.

Retest plan

Finding	Retest method	When
F1	Make failed logins from a test host and confirm the source is banned. Check that no accepted login exists for the source.	After the fix, then monthly
F2	Rerun Lynis and check that PKGS-7392 is gone.	After patching, then monthly
F3	Check service status, then send a test event and confirm it appears in the dashboard.	After the fix, then daily for one week
F4, F5	Confirm log age on disk and in the central store. Run chronyc tracking.	After the fix, then weekly
F6, F7	Rerun Lynis and compare the hardening index with 64.	After each fix, then quarterly
All	Internal audit checks a sample of the evidence and the saved script results.	Quarterly